

Deepali Banka

+91-8595998236 | deepalibanka@gmail.com | [linkedin.com/in/deepalibanka](https://www.linkedin.com/in/deepalibanka) | github.com/deepalib1710

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Electronics and Computer Engineering | GPA: 8.9/10

Chennai, India
Aug 2023 – Jun 2027

TECHNICAL SKILLS

Core Research Paradigms: Signal Processing (SP), Cyber-Physical Systems (CPS), Computer Vision (CV), Real-Time Embedded Systems, Acoustic Deep Learning (Acoustic AI), Defense R&D

Algorithmic and Machine Intelligence: Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), Sensor Array Fusion, Time-Difference-of-Arrival (TDOA), Statistical Signal Estimation, Multi-Sensor Synchronization (NTP)

Languages and Frameworks: Lingua Franca (LF), Python, C/C++, Java, SQL, MATLAB, PyTorch, OpenCV, Scikit-learn, NumPy, Pandas, Seaborn, Git, Linux/Unix

RESEARCH AND PROFESSIONAL EXPERIENCE

Indian Institute of Technology Delhi (IIT Delhi)

Research Intern, Centre of Applied Research in Electronics (Advisor: Prof Arun Kumar)

New Delhi, India
May 2026 – Till Date

- Conducting core defense research to develop real-time Signal Processing architectures and Deep Neural Network (DNN) algorithms for transient detection and spatial direction-finding of gunshots.
- Utilizing Time-Difference-of-Arrival (TDOA) matrices to evaluate wave segregation models (muzzle-blast vs. shock-wave).
- Curating acoustic datasets and implementing sensor array fusion to improve DNN resilience against false alarms in noisy environments.

Nanyang Technological University (NTU)

Research Intern, College of Computing and Data Science (Advisor: Prof Anupam Chattopadhyay)

Singapore | Remote
Aug 2025 – Till Date

- Conducting advanced systems research to evaluate deterministic execution and temporal semantics in distributed Cyber-Physical Systems (CPS) by implementing Network Time Protocol (NTP)-based time-synchronization frameworks.
- Developing low-latency Reactor models inside Lingua Franca to guarantee microsecond-level precise timing and eliminate race conditions for safe parallelism across multicore and distributed compute architectures.

Samsung R&D Institute

Student Researcher

Bangalore, India
Jul 2025 – Jun 2026

- Conducted Machine Learning research focused on acoustic de-reverberation methods to remove echo artifacts from corrupted speech and music signals.
- Built data synthesis pipelines to isolate clean time-series audio parameters from highly reverberant ambient noise.

Cadence Design Systems

Summer Intern

Noida, India
May 2025 – Jun 2025

- Designed and developed a graph-based pattern matching algorithm to automatically recognize recurring patterns in the design.
- Implemented cursor-centric Zoom and Pan features in Allegro System Capture.

Team MOVIS (Build custom Drones and Rovers, annual participation in NASA HERC)

Technical Team Member

Chennai, India
May 2024 – Jun 2025

- Developed an image stitching program using Python OpenCV to merge images by detecting and matching their feature points.
- Simulated the navigation of a GPS-Denied Autonomous Drone on Martian surface for ISRO Robotics Challenge, 2025.

School of Electronics Engineering (SENSE), VIT

Summer Intern

Hybrid
May 2024 – Jul 2024

- Summer Internship in VLSI Design, offered by the Centre for Nanoelectronics and VLSI Design (CNVD).
- Worked on the design and ASIC implementation of "Fetch Unit" for a pipelined RISC-V processor [[Paper](#)].

PUBLICATIONS AND MANUSCRIPTS

- Deepali Banka and Anupam Chattopadhyay, "From Imperative to Reactive Time-Synchronized Systems: An NTP Case Study" [in preparation].

PROJECTS

Edge-Accelerated Malware Detection via Hybrid FPGA-CPU Deep Learning Pipeline

Deployed quantized CNNs onto a PYNQ-Z2 FPGA using Quantization-Aware Training (QAT) and fused INT8 kernels to achieve low power, high-speed malware inference at the edge.

Human Fall Detection and Risk Prediction

Trained YOLOv8m architecture on a custom-curated vision dataset to predict human fall trajectories and risk parameters.

Image Processing and Canny Edge Detection

Implemented pixel-level Canny edge detection algorithms and enhancement filters to optimize visual feature extraction pipelines.

ACADEMIC ACHIEVEMENTS AND LEADERSHIP

- Led my team to win the Triad Battle '25, secured runners-up in AWS DeepRacer '23, and finished as a finalist in the VITISH '23 hackathon
- Among top 5% participants in Adobe GenSolve, '24
- Secured near perfect scores in C, C++ and Python programming tests conducted by Indian Institute of Technology, Bombay, '24
- Student fellowship, VLSI Design and Test Conference (VDAT), '24

COMMUNITY SERVICE AND CO-CURRICULAR

- Certified Senior First-Aid Professional (St. John Ambulance and Indian Red Cross Society) for emergency response management; delivered community programs and voluntary charitable support for underprivileged youth via Aap Ka Prayas NGO and Smile Foundation.
- Long-term training in Indian Classical Dance, won district and state-level dance competitions with multiple competitive choreography awards.